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Hydrocarbon condensate detection and control

Abstract

Hydrocarbon condensate detection and control. The hydrocarbon condensate detection and control system includes detecting hydrocarbon in an aqueous mixture and controlling the flow of the aqueous mixture based on the hydrocarbon content of the aqueous mixture.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. Utility patent application Ser. No. 17/410,560, filed Aug. 24, 2021, which is a continuation-in-part of U.S. Utility patent application Ser. No. 17/091,800, filed Nov. 6, 2020, which claims priority to U.S. Provisional Patent Application No. 62/978,015, filed Feb. 18, 2020, and which claims priority to U.S. Provisional Patent Application No. 63/034,945, filed Jun. 4, 2020, and which is a continuation-in-part of U.S. Utility patent application Ser. No. 17/024,673, filed Sep. 17, 2020, which claims priority to U.S. Provisional Patent Application No. 63/022,351, filed May 8, 2020. The aforementioned U.S. Utility patent application Ser. No. 17/410,560 is also a continuation-in-part of the aforementioned U.S. Utility patent application Ser. No. 17/024,673, filed Sep. 17, 2020. All of the aforementioned applications are incorporated herein in their entireties.

BACKGROUND

Field of the Disclosure

(1) The present disclosure relates generally to detection of condensed hydrocarbons and particularly detecting hydrocarbon condensate in impaired fluids from oil and gas fracturing.

Description of the Related Art

(2) Natural gas production methods produce a variety of constituent materials, often including hydrocarbons that condense into liquids under ambient conditions. Those condensed hydrocarbons, often referred to as condensates in oil and gas field vernacular, pose a danger to handlers who are not aware of their presence because such hydrocarbon condensates can easily burn or explode when they exist in significant quantities. Hydrocarbon condensates are also valuable for use in various chemical processes.

(3) A significant safety concern is condensate accumulation in fracking water. A ramification of undetected condensate impinging upon fracking sites and trucking assets used in fracking operations is the danger of explosion. Early catastrophic accidents experienced in basins across the United States can be directly correlated to condensate encroachment—that is, condensate ended up being in areas where it never should have been.

(4) Condensate can be produced as part of a well's natural lifecycle throughout the production phase of natural gas wells, particularly in wet gas producing wells. Production fluid gathering systems are generally constructed on well locations to store both production water and condensate that is gathered through the normal production cycle. The production water and hydrocarbons striate within the production tank infrastructure to allow for water haulers to haul production water for reuse or disposal, and hazardous material haulers to haul condensate to separation facilities for marketing.

(5) While operational procedures can be implemented with a goal to prevent the hauling of flammable liquids by non-sanctioned haulers, errors including those relating to faulty telemetry or operational oversight can lead to potentially unsafe conditions. In January of 2003, a truck fire erupted that resulted in three people being seriously burned and two people being killed. The cause of the fire was determined to be condensate contamination of non-flammable liquid that was caused by the release of hydrocarbon vapor during the unloading process. This accident could have been prevented if the condensate were observable and detected before it was improperly released. There are many other examples of dangerous incidents that have occurred in the past, and these types of incidents continue to have catastrophic potential across gas producing basins.

(6) The problem of condensate migration into unwanted operational areas is further compounded when production water is reused for completion operations. Centralized water offload locations designed to reuse water from well sites create a more functional platform for operational water reuse. Troublingly, such water reuse operations create opportunities for production water combined with condensate to be offloaded into centralized facilities, contaminating the bulk water and requiring costly cleanup efforts.

(7) Accordingly, there is a need to detect hydrocarbon condensate.

(8) There is also a need to prevent hydrocarbon condensate from being transferred to tanks and other vessels and locations that are not equipped to handle hydrocarbon condensate.

(9) There is also a need to provide a warning when there is a dangerous level of hydrocarbon

condensate in water or an aqueous mixture of water and other substances, such as production water used in a drilling site.
(10) There is also a need to separately collect hydrocarbon condensate that is present with other materials. That hydrocarbon condensate may be collected to improve the safety of the other material, to make the hydrocarbon condensate useful, or both.
(11) Accordingly, the present invention provides solutions to the shortcomings of storing and hauling substances that may contain hydrocarbons that may condense out of the substances and provides detection of such hydrocarbons. Those of ordinary skill in the art will readily appreciate, therefore, that those and other details, features, and advantages of the present invention will become further apparent in the following detailed description of the preferred embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above-mentioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and the disclosure itself will be better understood by reference to the following descriptions of embodiments of the disclosure taken in conjunction with the accompanying drawings, wherein:
(2) FIG. **1** illustrates an embodiment of a hydrocarbon detection and control system;
(3) FIG. **2** illustrates an embodiment of an aqueous fluid transfer system employing a hydrocarbon detection and control system;
(4) FIG. **3** illustrates another embodiment of an aqueous fluid transfer system employing a hydrocarbon detection and control system;
(5) FIG. **4** illustrates an embodiment of a plurality of hydrocarbon condensate detection systems deployed in a manifold pipeline fluid transfer system;
(6) FIG. **5** illustrates an embodiment of a processor-based controller for use in a hydrocarbon condensate detection system; and
(7) FIG. **6** illustrates an embodiment of a method of detecting hydrocarbon condensate.
(8) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate exemplary aspects of the disclosure, and such exemplifications are not to be construed as limiting the scope of the disclosure in any manner.
SUMMARY OF THE INVENTION
(9) In an embodiment, a hydrocarbon detection and control system for an aqueous mixture includes a controller, a hydrocarbon sensor, and a control device. The controller has at least one input to which the hydrocarbon sensor is coupled and the controller has at least one output to which the control device is coupled. The hydrocarbon sensor senses presence of hydrocarbon in an aqueous mixture and provides a signal to the controller commensurate with an amount of hydrocarbon in the aqueous mixture. The control device modifies flow of the aqueous mixture.
(10) In another embodiment, a multi-input pipeline hydrocarbon detection and control system is provided. The multi-input pipeline hydrocarbon detection and control system includes a plurality of pipelines, each to receive aqueous process fluid from a single separate unloading station and discharge that aqueous process fluid into a common manifold pipeline. Each unloading station has a processor that contains instructions which, when executed by the processor, cause the processor to receive a hydrocarbon content signal at a first input, the hydrocarbon sensor to sense hydrocarbon content of an aqueous process fluid and the hydrocarbon sensor signal indicating an amount of hydrocarbon content in the aqueous process fluid; and transmit a control signal to a control device coupled to an output, the control device to modify flow of the aqueous process fluid.
(11) A method of detecting hydrocarbon in an aqueous mixture and controlling flow of that aqueous

mixture is also provided. That method includes receiving a hydrocarbon content signal from a hydrocarbon content sensor disposed in an aqueous mixture, calculating hydrocarbon content of the aqueous mixture from the sensed hydrocarbon content sensor; and controlling the flow of the aqueous mixture based on the hydrocarbon content of the aqueous mixture.

(12) Other embodiments, which may include one or more portions of the aforementioned apparatuses and methods or other parts or elements, are also contemplated, and may have a broader or different scope than the aforementioned apparatuses and methods. Thus, the embodiments in this Summary of the Invention are mere examples, and are not intended to limit or define the scope of the invention or claims.

DETAILED DESCRIPTION

(13) The following description is provided to enable those skilled in the art to make and use the described embodiments contemplated for carrying out the concept. Various modifications, equivalents, variations, and alternatives, however, will remain readily apparent to those skilled in the art. Any and all such modifications, variations, equivalents, and alternatives are intended to fall within the spirit and scope of the present concept.

(14) Any reference in the specification to “one embodiment,” “a certain embodiment,” or a similar reference to an embodiment is intended to indicate that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of such terms in various places in the specification do not necessarily all refer to the same embodiment. References to “or” are furthermore intended as inclusive, so “or” may indicate one or another of the ored terms or more than one ored term.

(15) FIG. **1** illustrates an embodiment of a hydrocarbon condensate detection system **10**. The hydrocarbon detection system of FIG. **1** includes a fluorometer **12** to measure the concentration of hydrocarbons in a fluid, a turbidity sensor **14** to measure an extent of suspended solids in the fluid, a color sensor **16** to measure the color of the fluid, and a controller **600** to receive signals from the fluorometer **12**, turbidity sensor **14**, and color sensor **16** and to provide an output based on the presence of significant hydrocarbons in the fluid. The sensors **12**, **14**, and **16** may be disposed in a line **20** through which an aqueous mixture flows.

(16) The fluorometer **12**, turbidity sensor **14**, and color sensor **16** are coupled to inputs of a processor-based control device, such as the controller **600** illustrated in FIG. **5** or a microprocessor that may be packaged with the sensors **12**, **14**, **16** as a unit. The sensors provide a signal to the controller **600** that is commensurate with the amount of hydrocarbons in the aqueous mixture for the fluorometer **12**, commensurate with turbidity of the aqueous mixture for the turbidity sensor **14**, and commensurate with aqueous mixture color for the color sensor **16**. The controller **600** furthermore controls the flow of an aqueous mixture flowing past the fluorometer **12**, turbidity sensor **14**, and color sensor **16** through a flow control output, to modify flow of the aqueous mixture when hydrocarbon content of the aqueous mixture reaches one or more levels, such as a shutdown level that may be predetermined. A hysteresis may be provided to return to the original flow pattern when hydrocarbon content of the aqueous mixture returns to a desired level, which may be predetermined and may be offset from the shutdown level. The flow control output may be any desired output, for example an analog or digital signal that accomplishes flow control. The fluorometer **12**, turbidity sensor **14**, color sensor **16**, or a sensor package **28** comprising a combination of those sensors **12**, **14**, and **16** may also provide a 4 mA or other desired signal as a sensor operational output that may indicate that the sensors are working properly or indicate that one or more of the sensors has become dirty or is otherwise not properly sensing hydrocarbons.

(17) The aqueous mixture may be a mixture that is used in a shale fracturing or fracking operation and may be referred to as production water or impaired water. Fracking may refer to an operation employed to extract desired materials, such as natural gas or oil, from shale or other geological formations. Such a fracking production water aqueous mixture may include water, sand, and various compounds intended to aid in the flow of natural gas, oil, or other desired materials from

the shale.

(18) Hydrocarbons, such as those included in natural gas and oil, are generally separated from the aqueous mixture, but some hydrocarbons may bypass the separation process and can be mixed with or layered with the fracking aqueous mixture. Those hydrocarbons can be valuable when separated from the fracking aqueous mixture, but can be dangerous when found with the fracking aqueous mixture. For example, a tank of fracking aqueous mixture without hydrocarbons may be safely handled or transported and may be used in another fracking site or other operation, while a tank that contains hydrocarbons with the aqueous mixture may be a fire or explosion danger, particularly when it is unknown that the tank includes hydrocarbons and is not handled with the care to be taken for a tank containing hydrocarbons.

(19) Hydrocarbons generally have a different fluorescence than water or an aqueous fracking mixture. Accordingly, the fluorometer **12** of this invention can be used to sense hydrocarbons in water and aqueous mixtures, including fracking aqueous mixtures. In an embodiment in which hydrocarbons are to be sensed in an aqueous mixture, the fluorometer is arranged to sense hydrocarbon fluorescence. A level of hydrocarbon content may be set in the control device (e.g., **600**) and if hydrocarbon fluorescence reaches a predetermined level, as sensed by the fluorometer **12** in the aqueous mixture, the controller (e.g., **600**) may initiate an action, such as providing an alarm to an operator or attendant, closing a valve through which the aqueous mixture is flowing, or diverting flow of the aqueous mixture to a different tank or location. Hydrocarbon sensing may, for example, be performed in units of parts per million, parts per billion, or another level of hydrocarbon molecules or units suspended in the fluid or an amount of hydrocarbon present per unit of aqueous mixture.

(20) Hydrocarbons may fluoresce differently in fluid of certain colors and in fluids containing certain suspended solids or quantities of suspended solids. Accordingly, turbidity, or a quantity of suspended solids in the fluid, and color of the fluid, are sensed in certain hydrocarbon sensing embodiments disclosed herein and the resulting turbidity and color may be used to filter the sensed fluorescence to remove inaccuracies in sensed fluorescence caused by turbidity and color, and thereby to acquire a more accurate fluorescence and a more accurate determination of hydrocarbon content of the fluid.

(21) Further with regard to hydrocarbon fluorescence in turbid aqueous mixtures, hydrocarbons may not fluoresce as well or may fluoresce excessively in fluid with high turbidity, or with an excessive amount of solid matter carried in the fluid, as they would in clean water. To correct hydrocarbon sensing by fluorescence, a turbidity sensor may emit light in the red (i.e., 610-800 nm) or infrared (i.e., >800 nm) ranges and measure the scattering of light from suspended particles to determine turbidity of the aqueous mixture and as a filter to hydrocarbon content determination. Thus, the effect of turbidity on hydrocarbon fluorescence may be taken into account when determining hydrocarbon content of the aqueous mixture.

(22) Also, with regard to hydrocarbon fluorescence in colored aqueous mixtures, hydrocarbons may not fluoresce as well or may fluoresce excessively in fluid of certain colors. To correct hydrocarbon sensing by fluorescence, a color sensor may measure the color of the aqueous mixture and the hydrocarbon content determination may take the effect of color on hydrocarbon fluorescence into account when determining hydrocarbon content of the aqueous mixture.

(23) The sensors **12**, **14**, and **16** may be combined in a sensing package **28**, partially combined, or separate, and certain sensors (e.g., turbidity sensor **14** and color sensor **16**) may not be employed in certain circumstances where, for example, turbidity or color are not significant factors in hydrocarbon fluorescence. In many applications, however, the color and turbidity properties of the aqueous mixture are of interest in measuring or calculating concentrations of hydrocarbons in an aqueous mixture to compensate for fluctuations in the fluorescence measurement made by the fluorometer **12**. A sensor housing **18** may furthermore include the fluorescence sensor **12**, the turbidity sensor **14** and the color sensor **16** and additional components that are not used in a

fluorescence determination, but that are nonetheless incorporated into the single probe housing **18** to simplify installation or to assure that all sensing is taking place in the same vicinity.

(24) Hydrocarbon content may be sensed in a variety of locations, including pipelines, transfer hoses **64**, and tanks **212**, **214**, and **216**. Where hydrocarbon sensing is performed outdoors and the sensors **12**, **14**, **16**, sensor unit, **28**, or complete hydrocarbon detection and control system **10** is installed outdoors, protections may be provided to protect the outdoor equipment, including static protection and lightening protection.

(25) FIG. **2** illustrates an embodiment of an aqueous fluid transfer system **200** employing a hydrocarbon detection and control system **10** installed in or attached to a tanker truck **210** outlet **221**, a line **222** leaving the tanker truck **210** and entering one or more holding tanks **214** or **216**, or otherwise as desired to sense hydrocarbon mixed with the aqueous fluid. In that embodiment, an aqueous mixture is hauled to a drilling site or another site where the aqueous mixture is needed or to be stored. The aqueous mixture is hauled in the tank **212** of the tanker truck **210** in the embodiment illustrated in FIG. **2**. Transfer may begin with a transfer hose **64** acting as the line **222** extending from the truck tank **212** to the aqueous mixture holding tank **214** or **216**. The hydrocarbon detection system **10** may be installed in the line **222**, coupled to the truck **210** at the inlet of the line **222** or transfer hose **64**, or otherwise situated such that the aqueous mixture passes the hydrocarbon detection system **10** as it flows from the truck tank **212** such that the hydrocarbon detection system **10** senses hydrocarbon content in the aqueous mixture leaving the tank **212** of the tanker truck **210**.

(26) Aqueous mixture transfer from a truck **210** tank **212** typically begins with transfer of aqueous mixture **48** into the aqueous mixture tank **214**. If hydrocarbon content of the aqueous mixture leaving the truck tank **212** exceeds a predetermined limit, as sensed by the hydrocarbon detection system **10**, the hydrocarbon detection system **10** takes action to stop transfer of the aqueous mixture **48** to the aqueous mixture holding tank **214**. For example, in embodiments, if a hydrocarbon content of the transferring aqueous mixture **48** being sensed by the hydrocarbon detection system **10** exceeds 800 ppm, the hydrocarbon detection system stops flow of the aqueous mixture **48** by de-energizing a pump **22**, closing a valve **60**, actuating a diverting valve **318**, de-energizing a compressor providing pressure to the truck tank **212**, or de-energizing a vacuum creating device that draws the aqueous mixture from the tank **212**. Alternatively, the hydrocarbon detection system **10** may provide an audible or visual alarm **40** or otherwise notify an operator that the hydrocarbon level of the aqueous mixture **48** is too high for transfer to continue into the aqueous mixture holding tank **214** or that hydrocarbon content of the aqueous mixture **48** is otherwise out of range. The hydrocarbon detection system **10** or an operator may alternatively, or in addition, divert the aqueous mixture **48**, redirecting the aqueous mixture from, for example an aqueous mixture reservoir, to another location, such as a hydrocarbon and aqueous mixture reservoir or a hydrocarbon condensate tank **216** when excess hydrocarbon content is sensed in the aqueous mixture **48**. For example, the hydrocarbon detection system **10** may divert the flow of aqueous mixture **48** by actuating a diverting valve **318** or one or more two-way valves, such that the aqueous mixture **48** is transferred to the condensate tank **216** rather than the aqueous mixture tank **214**. Where no diverting valve **318** is employed, the flow of aqueous mixture **48** from the truck **210** tank **212** to the aqueous mixture holding tank **214** may be terminated and the line **222** or transfer hose **64** may be moved to the hydrocarbon condensate tank **216** such that transfer of aqueous mixture from the truck **210** tank **212** may continue.

(27) Valves **60**, **318** may include fast acting pneumatic or electric valves that, upon detection of condensate levels of a predetermined hazardous PPM range, can quickly stop or redirect flow. Redirection of the flow of aqueous mixtures containing excess hydrocarbons can allow for the capture and separation of hydrocarbon condensate in one or more separate process vessels to eliminate safety hazards while allowing for the resale of the captured hydrocarbons.

(28) In an embodiment of the hydrocarbon detection system **200** of FIG. **2**, an operator pulls a truck

210 into the vicinity of a fracking water, aqueous mixture tank **214**. Information about the truck **210** is communicated to a site controller, either automatically or upon initiation by the driver or an operator. An operator then connects the tank **212** of the tanker truck **210** to a fracking water aqueous mixture tank **214** for transfer of aqueous mixture.

(29) FIG. **3** illustrates an embodiment 300 in which aqueous mixture transfer from an aqueous mixture tank **214** to a truck **210** tank **212** may be accomplished using the hydrocarbon detection system **10**. In that embodiment, transfer of aqueous mixture to the truck **210** tank **212** is halted if a hydrocarbon content higher than desired is sensed in aqueous mixture being transferred to the truck **210** tank **212** using any of the processes described in connection with the aqueous mixture transfer described in connection with FIG. **2**. In such an embodiment, the hydrocarbon detection system **10** may be placed anywhere in the line **222**, whether the line **222** is a transfer hose **64**, solid pipe, a manifold system, or a combination of such elements. The hydrocarbon detection system **10** may, for example, be located in an aqueous mixture tank **214** outlet **302** that leads to the truck **210** tank **212**. If the hydrocarbon detection system **10** senses a level of hydrocarbons above its predefined setting or limit, truck **210** tank **212** loading will be stopped by valve, pump de-energization, operator, or as described elsewhere herein. High hydrocarbon (also referred to as hydrocarbon condensate or simply condensate) content aqueous mixture may be diverted to a condensate tank **216** or another location. Hydrocarbons may further be removed from the high hydrocarbon aqueous mixture for sale or other use and the aqueous mixture may be used after the hydrocarbons are removed.

(30) In an embodiment, aqueous mixture containing a high hydrocarbon content may be left to settle and separate in the condensate tank **216**. When settled, the hydrocarbons will generally separate from the aqueous mixture with hydrocarbons floating above or on top of the aqueous mixture. Once settled, the aqueous mixture may be drawn from the bottom of the condensate tank **216** and moved to a truck **210** tank **212** or aqueous mixture tank **214** for further use. Alternatively or in addition, the hydrocarbon condensate may be drawn from the hydrocarbon condensate tank **216** for use elsewhere.

(31) FIG. **4** illustrates an embodiment of a plurality of hydrocarbon condensate detection systems **10** deployed in a manifold pipeline fluid transfer system **100** through which an aqueous mixture flows and, in certain embodiments, is propelled. In the embodiment illustrated in FIG. **4**, one hydrocarbon detection system **10** of FIG. **1** is installed in each line **20** through which an aqueous process fluid flows. Those hydrocarbon detection systems **10** sense hydrocarbons flowing in the process fluid. In the embodiment illustrated, two unloading stations **110** are provided, though any number of stations desired may be included in a manifold pipeline fluid transfer system **100**. A truck **210** tank **212** may be coupled to each unloading station **110**, for example, by way of a separate transfer hose **64** for each truck **210** tank **212**. Each unloading station **110** includes a hydrocarbon sensing package **28** and controller **600** that may include a combined fluorescence, turbidity, and color sensor hydrocarbon detection system **10**. Each unloading station **110** may also include a variety of sensors, valves, gauges, and displays, such as an offloading valve **140**, a sight tube **142**, a pressure gauge or sensor **144**, a flow gauge or sensor **146**, a temperature gauge or sensor **148**, or any other desired gauge or sensor. For the purpose of the embodiment illustrated in FIG. **4** gauges may provide a visual representation of a characteristic of the fluid in the offloading portion of the line **20** or pipe, and a sensor or sensors may provide signals that represent a fluid characteristic to a processor-based device, such as the controller **600** illustrated and discussed in connection with FIG. **5** or a control system or other computing device. Sensor signals may, furthermore, be transmitted through wires or wirelessly.

(32) The manifold pipeline fluid transfer system **100** of this embodiment may be employed to expedite offloading of tanker trucks **210** while stopping unloading when high or dangerous hydrocarbon content is sensed in the aqueous mixture being unloaded. In the embodiment illustrated in FIG. **4**, a plurality of tanker truck **210** offloading stations **110** are provided so that

more than one tanker truck **210** can have fluid from their associated tanks **212** offloaded simultaneously into the manifold. For example, in an embodiment for offloading tanks of water-based fluids used in hydraulic fracturing, ten offloading stations **110** may be provided. A tanker truck **210** may pull up to and connect to the manifold pipeline fluid transfer system **100** through an offloading station **110** when that offloading station **110** is not occupied or otherwise used by another truck **210**. An offloading station **110** can be used independent of other station occupancy, regardless of any use or maintenance occurring of the other offloading stations **110**, and regardless of whether one or more trucks **210** occupying one or more other stations are in the process of offloading. The tanker truck **210** that pulls into an offloading station can connect to an offloading portion of the line **20** at the offloading station **110** and offload the contents of its tank **212** through the offloading portion of the line **20** or pipe while other trucks **210** are simultaneously offloading their tanks **212**.

(33) In certain embodiments, each offloading station **110** includes a flexible transfer hose hook-up **152** through which fluid from the tank **212** of the truck **210** can flow into a manifold **114** and from there, directly or indirectly, into a site tank **120**. The manifold **114** may be in fluid communication with each offloading station **110** and the site tank **120**. The manifold **114** may be a piping system that accepts fluid from each offloading station **110**, combines fluid received from the offloading stations **110** at one or more junctions **154**, and deposits the fluid received from the offloading stations **110** into the site tank **120**.

(34) The manifold pipeline fluid transfer system **100** may be configured in various ways that are suited to an offloading site. The manifold pipeline fluid transfer system **100** illustrated in FIG. **4** includes a plurality of lines **20** that flow through offloading stations **110** into a manifold **114** branch extending to each offloading station **110**. The plurality of lines **20** extending from the offloading stations join at one or more junctions **154** to a common manifold **114** that extends to a tank such as an aqueous mixture storage tank **120** illustrated in FIG. **4**. The common line **156** may alternatively extend to any desired destination into which the fluid from the truck **210** tanks **212** is desired to be deposited, including, for example, a storage vessel of any type, an underground storage area, a well, or a rig of some type, such as a drill rig.

(35) A flow sensor or meter **150** or another flow measuring device may be placed in the manifold **114** common line to measure the total flow through the manifold **114**. That flow measurement may be used to determine total flow into the manifold **114** and may be used to control flow into the manifold **114**, for example, through the offloading valves **140**. Alternatively, a flow switch may be placed in the manifold **114** common line to indicate fluid is flowing through the manifold **114**. Either the flow meter **150** or switch may be coupled to a computerized monitoring or control system including the processor-based control device **600** illustrated and discussed in connection with FIG. **5**.

(36) The offloading valves **140** may be located in each offloading station **110** to control or flow from the offloading stations **110** or to isolate one or more offloading stations **110**, for example when high hydrocarbon content is sensed in the aqueous mixture flowing into that offloading station **110**. The offloading valves **140** may be a variety of types of valves, including a ball valve, a gate valve, or a globe valve. The offloading valve **140** may furthermore be actuated manually, may be automatically controlled for full opening or closure, or may be modulated for regulated flow by a manifold control system, such as the controller **600** illustrated in FIG. **5**. The offloading valve **140** may be of a size that permits full flow of fluid from the truck **210** tank **212**, such as A 4″ valve with a full flow characteristic. The full flow characteristic of a ball valve, for example, may be beneficial to enable fast offloading of tanks **212** coupled thereto. Alternatively, the offloading valve **140** may be configured with a linear control characteristic, such as that provided by a globe valve, or may have another desired flow characteristic. The offloading valve may furthermore be used to permit flow through the offloading station **110** when open and to prevent flow into the offloading station **10** when closed.

(37) Control of the offloading valve **140** and other components of the manifold system **100** may be performed using a computer, such as the processor-based controller **600** illustrated and discussed herein in connection with FIG. **5**.
(38) Fluid unloaded from a tanker truck **210** is known to sometimes encounter gas, particularly air, mixing with the fluid. Thus, an air removal system **160** may be provided in the manifold system **100**.
(39) In an embodiment in which an offloading valve **140** is included in the unloading station **110**, the offloading valve **140** is opened when the transfer hose **64** is attached to the unloading station **110**. The hydrocarbon detection and control system **10** is coupled to an actuator **141** on the offloading valve **140** and the actuator **141** closes the offloading valve **140** when a hydrocarbon limit set in the hydrocarbon detection and control system **10** exceeds its preset limit, which may be 800 ppm of hydrocarbons in the aqueous mixture flowing past the hydrocarbon detection and control system **10**.
(40) It may be seen that in a manifold pipeline fluid transfer system **100** having multiple stations, while not necessary, it may be beneficial to have a separate hydrocarbon detection and control system **10** and offloading valve **140** at each station so that when the hydrocarbon limit is exceeded at one station, aqueous mixture flow through that station may be stopped while trucks **210** may continue to offload aqueous mixture at the remaining stations.
(41) In operation, the sensors **12**, **14**, and **16** portion of a hydrocarbon detection and control module **10** may be installed in a line, such as a pipe carrying the fluid for which hydrocarbon detection is desired. Installation of the sensors may be by insertion of the sensors **12**, **14**, and **16** into the line **20** and affixing the sensors **12**, **14**, and **16** in the line **20**. A pump **22** may propel the fluid through the line **20**, or the fluid may flow by way of gravity or other means. A flow sensor **150** may also be included in the line **20** and a pressure sensor **144** or other sensors may also or alternatively be included in the line **20**.
(42) Flow shut-off devices may be included in the line **20**, including a pump shutoff control and one or more valves, which may be used to isolate a section of the line **20** into which one or more of the sensors **12**, **14**, and **16** are inserted.
(43) The hydrocarbon detection and control module **10** may have its sensors **12**, **14**, and **16** encapsulated in a probe housing **18** that may be inserted into the line **20**. The hydrocarbon detection and control module **10** may provide continuous monitoring of the fluid stream and may have an accuracy of a single PPM of hydrocarbon.
(44) The hydrocarbon detection and control system **10** may be housed in a Class1/Div2 or Class1/Div1 enclosure for deployment in hazardous areas or another desired enclosure, and the hydrocarbon detection and control system **10** may be outfitted with a self-cleaning function tailored to the water stream to ensure that it does not lose detection ability due to plating or fouling. The hydrocarbon detection and control system **10** may be powered by a nominal 110 VAC power supply or another desired electrical power supply.
(45) The hydrocarbon detection and control system **10** may also include a sensor cleaning system to remove hydrocarbons, solid impurities, or other foreign materials from the fluorescence, color, and turbidity sensors **12**, **14** and **16**. In an embodiment, that cleaning system may draw **1**H P of power and create 7CFM of airflow across the sensors **12**, **14**, and **16** at 90 psi per hour.
(46) In certain embodiments, the hydrocarbon detection and control system **10** may include cellular or satellite connectivity so that the system can be accessed from anywhere in the world allowing for visibility into condensate concentrations and alarm conditions.
(47) Regarding identification provided when a tanker truck **210** approaches a site, such as an offloading site or an onloading site, the truck **210** or tank **212** may be identified by any unique identifier of the tank **212** or the truck **210** on which a particular tank **212** is mounted and may be recognized in a variety of ways. For example, a user interface may be used to identify the tank **212** currently in position to operate (e.g., load or unload) by way of a wired or wireless transmission

from an electronic device associated with the tank **212**. A unique identifier may be transmitted from the tank **212** or associated truck **210** by any signal transmitting device, or an identifier may be read and transmitted by a geofencing or other position determination device that senses the presence of the tank **212** or its associated truck **210**. Alternatively, a driver or operator may enter the tank identifier into a device to recognize a tank **212** that is currently under operation.

(48) In an embodiment, a tanker truck driver will carry an electronic fob **36** or other device that contains a name or an identification of the driver, the company that operates the truck **210**, a truck **210** identifying number, or any other tanker truck **210** related information desired, such as a quantity of aqueous mixture carried by the truck tank **212** or a capacity of the tank **212**, and transmits that identification to an appropriate communicating node at, or remote from, the site.

(49) FIG. **5** illustrates an embodiment of a processor-based controller **600** for use in a hydrocarbon condensate detection system. The processor-based controller **600** may, for example, be a microcontroller, an application specific integrated circuit, a general-purpose computer, or a programable logic controller, such as those manufactured by Rockwell International under the Allen Bradley trademark. In the embodiment illustrated in FIG. **5**, the processor-based device **600** includes a processor **602** and a communication device **612**. The processor **602** and communication device **612** can be combined in a microprocessor or other device and other components (e.g., **604** and **606**) may also be included in such a microprocessor or other device.

(50) The communication device **612** may be wired to a device to which it communicates. The communication device **612** may, in addition or alternatively, wirelessly communicate with one or more other devices over a network **240**, which may be a wireless network, such as a mobile smartphone network, a short-range wireless protocol, such as Bluetooth or Zigbee, or another desired communication protocol, such and the communication device **612** may operate both wired and wirelessly. The processor-based device **600** may furthermore include memory **604**, an input **610** that may receive an input signal, such as a signal transmitted by a sensor, and an output **608** that may transmit a control signal, instruction, or data to another device, such as a valve actuator or other controlled device. The output device may alternatively or in addition provide a reading, for example a current quantity of hydrocarbons present in the aqueous mixture flowing past one or more hydrocarbon sensors.

(51) The processor-based device **600** may also be coupled to a user interface **632** to receive one or more signals from, for example, one or more of a keyboard, touch screen, mouse, microphone or other input device or technology and may have associated software. The user interface **632** may also transmit information to, for example, a printer or screen coupled to the user interface **632** or the output **608**.

(52) The memory **604** may, for example, include random-access memory (RAM), flash RAM, dynamic RAM, or read only memory (ROM) (e.g., programmable ROM, erasable programmable ROM, or electronically erasable programmable ROM) and may store computer program instructions and information. In embodiments, the memory **604** may be partitioned into sections including an operating system partition **616** where system operating instructions are stored, a data partition **618** in which data, such as total hydrocarbon passage and total aqueous mixture passage may be stored, and a signal modification module **620** that may be used to convert sensor signals into human readable engineering units.

(53) The storage device **606** may include a memory device or a data storage device or a combination of both memory and data storage devices, or another device or devices for storage of data. The data storage **606** may be considered local storage when the data is stored directly on the processor-based device **600** or the data may be accessible to the processor-based device **606** over a wired or a wireless network. The storage device **606** may furthermore include a computer readable storage medium that includes code executable by the processor **602**, which may be used, for example, to cause the processor **602** to, at least in part, perform hydrocarbon detection as disclosed herein.

(54) In an embodiment, the storage device **606** for the processor-based device **600** may include a combination of flash storage and RAM. The storage device **606** may also include a computer readable storage medium and may include code executable by the processor **602**.
(55) In an embodiment, the elements, including the processor **602**, communication adaptor **612**, memory **604**, input device **610**, output device **608**, and data storage device **606** may communicate by way of one or more communication busses **614**. Those busses **614** may include, for example, a system bus or a peripheral component interface bus.
(56) The processor **602** may be any desired processor and may be a part of a controller **600** or a microcontroller, may be part of or incorporated into another device, or may be a separate device. The processor **602** may, for example, be an Intel® manufactured processor or another processor manufactured by, for example, AMD®, DEC®, or Oracle®. The processor **602** may furthermore execute the program instructions and process the data stored in the memory **604**. In one embodiment, the instructions are stored in the memory **604** in a compressed or encrypted format. As used herein the phrase, “executed by a processor,” is intended to encompass instructions stored in a compressed or encrypted format, as well as instructions that may be compiled or installed by an installer before being executed by the processor **602**.
(57) The data storage device **606** may be, for example, non-volatile battery backed static random-access memory (RAM), a magnetic disk (e.g., hard drive), optical disk (e.g., CD-ROM) or any other device or signal that can store digital information. The data storage device **606** may furthermore have an associated real-time clock, which may be associated with the data storage device **606** directly or through the processor **602**. The real-time clock may trigger data from the data storage device **606** to be sent to the processor **602**, for example, when the processor **602** polls the data storage device **606**. Data from the data storage device **606** that is to be sent across the network **622** through the processor **602** may be sent in the form of messages in packets if desired. Those messages may furthermore be queued in or by the processor **602**.
(58) The communication adaptor **612** permits communication between the processor-based device **600** and other nodes, such as a tanker truck controller **35** or a remote monitoring peripheral computer **37**, both illustrated in FIGS. **2** and **3**, or another computing device or server. The communication adaptor **612** may be a network interface that transfers information from a node such as a networked device, which would include an actuating device such as valve **60** or a sensing device **12**,**14**,**16**, to the tanker truck controller **35**, the remote monitoring peripheral computer **37**, a general-purpose computer (not illustrated), a user interface **632**, or another node. The communication adaptor **612** may be an Ethernet adaptor or another adaptor for another type of network communication. It will be recognized that the processor-based device **600** may alternatively or in addition be coupled directly to one or more other devices through one or more input/output adaptors (not shown).
(59) The processor-based control device **600** may communicate and may have its memory **604** or processes modified by a user interface **632**. That user interface **632** may be, for example, a computer; a tablet; a mobile smartphone device (referred to herein as a phone); an application specific user interface device; or another device that can be used to transfer information to the controller **600** or receive information from the controller **600**.
(60) FIG. **6** illustrates an embodiment of a method of detecting hydrocarbon condensate in an aqueous mixture **500**. At **502**, a processor **602**, which may be associated with the fluorometer, turbidity sensor, and color sensor or external to those sensors or a controller **600** receives a fluorometer signal from a fluorometer **12**, at **504** the processor **602** or controller **600** receives a turbidity signal from a turbidity sensor **14**, and at **506** the controller **600** receives a color signal from a color sensor **16**. At **508**, the processor **602** or controller **600** determines an amount of hydrocarbon content in the aqueous mixture from the fluorescence, the turbidity, and the color of the aqueous mixture. At **510**, the processor **602** or controller **600** actuates a control device to modify the flow of the aqueous mixture based on the hydrocarbon content of the aqueous mixture.

(61) The processor **602** or controller **600** may calculate the hydrocarbon content of the aqueous mixture based on the fluorometer signal, the turbidity sensor, and the color sensor. In certain embodiments, the hydrocarbon content may be determined based on the fluorescence of the aqueous mixture with sensed turbidity and color used to correct hydrocarbon content inaccuracies caused by turbidity and color components of the aqueous mixture.
(62) Control of flow of the aqueous mixture may be accomplished in various ways, including closing or switching of a valve **140**, **318** and by energizing or de-energizing a pump, compressor, or other component used in the transfer of the aqueous mixture.
(63) A truck tank pressurization system **25** may be provided to provide pressure or vacuum to a truck **210** tank **212**. The truck tank pressurization system **25** may be contained within the tanker truck and the truck tank pressurization system **25** may be powered by the tanker truck **210**. Alternatively, the truck tank pressurization system **25** may be external to the truck **210**, for example at an onloading or offloading site. The truck pressurization system **25** may furthermore be powered externally to the truck, again for example at an onloading or offloading site.
(64) Truck **210** tanks **212** and the vessels they are loading from or unloading into may be pressurized to enhance that process, for example, using the truck tank pressurization system **25**. In certain embodiments, when a truck **210** tank **212** is unloading, the tank **212** is pressurized to assist in moving fluid out of the tank **212**. In another embodiment, a vessel the tank **212** is unloading into may create a vacuum or negative pressure to assist in drawing the fluid out of the tank **212**. In embodiments where the tank **212** is being loaded, a vessel providing fluid to the tank **212** may be pressurized to assist the fluid in moving from the vessel to the tank **212** or the tank **212** may draw a vacuum to assist in moving the fluid from the vessel to the tank **212**. In various embodiments, the truck **210** may continue to operate and draw a vacuum until the transfer hose **64** is empty to drain the fluid in the transfer hose **64** into the tank **212**. In certain embodiments, the pressure or vacuum may be modified as loading or unloading operations progress, for example, reducing the pressure in a tank **212** as the fluid level in the tank **212** is reduced or reducing the vacuum in the tank **212** as the tank fills. For example, pressure provided to a tank **212** during unloading may be reduced when the tank **212** approaches empty to reduce the amount of air missing with the unloading fluid. As has been mentioned, the operator may shut the production water valve **60** when a tank **212** is filling and full all but the volume of the transfer hose **64**, pressure and suction may be removed or de-energized at that time, and once the transfer hose **64** has been emptied into the tank **212**, the tank **212** should have a full load of fluid.
(65) The user interface **632** may communicate with the transmitter **600** wirelessly or by a wired connection, for example at the transmitter **600** input/output device coupling **608**. The user interface **632** may, for example, be a multi-function communication device, such as an Apple iPhone, a Samsung Galaxy wireless phone, another wireless phone device, a tablet device, or another computing device. The user interface **600** may require input of a password, user facial identification, or other validation to access information contained within the transmitter **600**. the user interface may, furthermore, both access data held in the transmitter and configure the transmitter.
(66) The hydrocarbon condensate detection system **10** may include or be housed in an enclosure that is sealed against entry of moisture and solid material and debris and may be surge protected to protect against electrical overload. Communications between the hydrocarbon condensate detection system **10** may furthermore be password protected with unique passwords. Those passwords may relate to a customer, organization, site, or otherwise as desired to restrict access to information to those having a legitimate interest in the information to be shared between devices.
(67) The hydrocarbon condensate detection system **10** may also include one or more indicators **634** that extend through its enclosure to provide operational information to an onlooker. For example, the indicators **634** may include one or more LEDs that indicate information such as operating status of the transmitter **600** and communication between the transmitter **600** and another node. The

indicators **634** may furthermore include one or more readouts that indicate, for example, the signal being received by the transmitter **600** or the signal being or to be transmitted by the transmitter **600**.
(68) While this disclosure has been described as having exemplary designs, the present disclosure can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

Claims

1. A production water hydrocarbon detection and control system for sand and water-based production water aqueous mixture in fluid communication with a tanker truck tank, comprising: a controller having at least one input and at least one output; a hydrocarbon sensor having at least one of a fluorometer, a turbidity sensor, and a color sensor in a pipe in fluid communication with the tanker truck tank and a production site production water aqueous mixture tank coupled to a first input of the controller, the hydrocarbon sensor sensing a level of presence of hydrocarbon in the aqueous mixture flowing between the tanker truck tank and the production site aqueous mixture tank and providing a signal to the controller commensurate with an amount of hydrocarbon in the aqueous mixture; and a control device coupled to the controller output, the control device modifying flow of the aqueous mixture to stop flow of aqueous mixture between the tanker truck tank and the production site tank.
2. The hydrocarbon detection and control system of claim 1, wherein the hydrocarbon detection and control system is attached to an outlet of the tanker truck tank to sense hydrocarbon content of aqueous mixture leaving the tanker truck tank.
3. The hydrocarbon detection and control system of claim 2, wherein the controller redirects the transfer of the aqueous mixture flowing from the tanker truck tank to the production site tank to a hydrocarbon and aqueous mixture reservoir.
4. The hydrocarbon detection and control system of claim 3, wherein the controller redirects the transfer of the aqueous mixture to the hydrocarbon and aqueous mixture reservoir when a high level of hydrocarbon content is detected by the hydrocarbon sensor by one of actuating at least one valve and by notifying an operator of the high hydrocarbon content of the aqueous mixture.
5. The hydrocarbon detection and control system of claim 4, further comprising: a transfer hose to extend between the tanker truck tank and the production site tank; and the valve being disposed to control flow of aqueous mixture though the transfer hose.
6. The hydrocarbon detection and control system of claim 2, further comprising identifying a truck unloading the aqueous mixture to the controller.
7. The hydrocarbon detection and control system of claim 1, wherein the hydrocarbon detection and control system is attached to the production site production water aqueous mixture tank to sense hydrocarbon content of aqueous mixture leaving the production site production water aqueous mixture tank.
8. The hydrocarbon detection and control system of claim 7, wherein the controller stops the transfer of the aqueous mixture.
9. The hydrocarbon detection and control system of claim 1, wherein the flow of aqueous mixture is stopped when a high hydrocarbon content is sensed in the aqueous fluid by one of closing a valve, de-energizing a pump, de-energizing a compressor, and by notifying an operator of the high hydrocarbon content sensed in of the aqueous mixture.
10. The hydrocarbon detection and control system of claim 1, wherein modifying flow of the aqueous mixture includes modifying flow on one of a plurality of aqueous mixture pipelines

feeding a common pipeline.

11. The hydrocarbon detection and control system of claim 1, wherein the flow of aqueous mixture between the tanker truck tank and the production site tank is stopped when the hydrocarbon sensor senses a predetermined level of presence of hydrocarbons in the aqueous mixture.

12. The hydrocarbon detection and control system of claim 11, wherein a valve is actuated to divert the flow of aqueous mixture when the predetermined level of hydrocarbons is reached.

13. A tanker truck multi-input pipeline hydrocarbon detection and control system, comprising: a plurality of pipelines each pipeline terminating at an unloading station for fluid communication with a tanker truck tank, each pipeline to receive aqueous sand and water based process fluid from a single separate unloading station and discharge that aqueous sand and water based process fluid into a common process fluid manifold pipeline, each tanker truck unloading station having a processor containing instructions which, when executed by the processor cause the processor to; receive a hydrocarbon content signal coupled to an input from at least one of a fluorometer, a turbidity sensor, and a color sensor installed in one of the plurality of pipelines, the hydrocarbon content signal to indicate an amount of hydrocarbon content in the aqueous process fluid flowing into the pipeline from a tanker truck tank; and transmit a control signal to a control device coupled to an output, the control device to modify flow of the aqueous process fluid and stop flow of aqueous process fluid flowing from the tanker truck tank into the common process fluid manifold when the hydrocarbon content of the aqueous process fluid exceeds a predetermined amount.

14. The multi-input pipeline hydrocarbon detection and control system of claim 13, wherein the control device to modify flow of the aqueous process fluid controls flow through a single offloading station.

15. The multi-input pipeline hydrocarbon detection and control system of claim 14, further comprising identifying a truck unloading the aqueous mixture to the controller.

16. The multi-input pipeline hydrocarbon detection and control system of claim 14, further comprising directing the aqueous sand and water based process fluid to a hydrocarbon and aqueous mixture reservoir when the predetermined level of hydrocarbons is reached.

17. A method of detecting hydrocarbon in an aqueous mixture flowing between a tanker truck tank and a production site tank and controlling flow of that aqueous mixture, comprising: receiving a hydrocarbon content signal from a hydrocarbon content sensor comprising at least one of a fluorometer, a turbidity sensor, and a color sensor in a pipe in fluid communication with the tanker truck tank and the production site tank disposed in an aqueous mixture; calculating hydrocarbon content of the aqueous mixture from the sensed hydrocarbon content sensor; and controlling the flow of the aqueous mixture to stop flow of the aqueous mixture between the tanker truck tank and the production site tank when based on the hydrocarbon content of the aqueous mixture exceeds a predetermined limit.

18. The method of claim 17, wherein flow of the aqueous mixture is stopped through only one of a plurality of unloading stations flowing into a common pipeline.

19. The method of claim 18 further comprising identifying a truck unloading the aqueous mixture the flow of which is stopped.

20. The method of claim 17, further comprising redirecting the transfer of the aqueous mixture from an aqueous mixture reservoir to a hydrocarbon and aqueous mixture reservoir.